AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

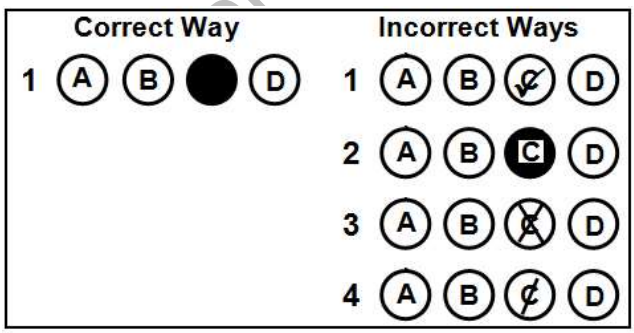
Chemistry Paper I

Time: 1 hour 30 minutes Marks: 50

AKU-EB Annual Examinations 2024 For Teaching & Learning only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. A scientist uses 52 g of Na_2SO_4 for a chemical reaction.

   How many moles of Na atoms does the compound contain?

   (**Note:** Molar mass of Na = 23 g mol^{-1}, S = 32 g mol^{-1} and O = 16 g mol^{-1})

   A. 0.366 mol
   B. 0.732 mol
   C. 8.422 mol
   D. 16.845 mol

2. If a mixture of 7 cm^3 of CH_4 and 7 cm^3 of C_2H_4 is burnt, then the volume of oxygen required for their complete combustion is

   A. 7 cm^3.
   B. 14 cm^3.
   C. 21 cm^3.
   D. 35 cm^3.

3. In PCl_5, the percentage of chlorine atoms by mass is

   (**Note**: $^{31}_{15}P$ and $^{35.5}_{17}Cl$)

   A. 35.5%.
   B. 85.13%.
   C. 177.5%.
   D. 208.5%.

4. The use of a balanced chemical equation to calculate the amount of product that will form during a chemical reaction gives the

   A. actual yield.
   B. fractional yield.
   C. theoretical yield.
   D. percentage yield.

5. The oxidation number of sulphur in SO_3^{2-} is

   A. + 2
   B. + 3
   C. + 4
   D. + 6

6. An example of disproportionation redox reaction in which the same element is oxidised as well as reduced is

   A. $2Na_2S_2O_{3(aq)} + I_{2(s)} \rightarrow Na_2S_4O_{6(aq)} + 2NaI_{(aq)}$
   B. $2FeCl_{3(aq)} + H_2S_{(g)} \rightarrow 2FeCl_{2(aq)} + 2HCl_{(aq)} + S_{(s)}$
   C. $P_{4(s)} + 3NaOH_{(aq)} + 3H_2O_{(l)} \rightarrow 3NaH_2PO_{2(aq)} + PH_{3(g)}$
   D. $2KMnO_{4(aq)} + 8H_2SO_{4(aq)} + 10FeSO_{4(aq)} \rightarrow K_2SO_{4(aq)} + 2MnSO_{4(aq)} + 5Fe_2(SO_4)_{3(aq)} + 8H_2O_{(l)}$

AKU-EB Annual Examinations 2024 for Teaching & Learning only

7. Consider the following redox reaction in an aqueous medium.

$Fe^{+2} + Ag^{+} \rightarrow Fe^{+3} + Ag$

The standard reduction potential values for iron and silver are +0.77 V and +0.80 V, respectively.

The overall cell potential and the feasibility of the reaction will be

	Overall Cell Potential	**Feasibility of the Reaction**
A	+1.57 volts	spontaneous
B	+0.03 volts	spontaneous
C	−0.03 volts	non-spontaneous
D	−1.57 volts	non-spontaneous

8. To convert from Coulombs to Faraday, the value must be

A. subtracted from 1.04×10^{-5}
B. multiplied by 1.04×10^{-5}
C. subtracted from 9.65×10^{4}
D. multiplied by 9.65×10^{4}

9. During the discovery of electrons, a glow is produced inside the discharge tube. The colour of glow depends on the

A. nature of gas.
B. volume of gas.
C. metal used as cathode.
D. pressure inside the discharge tube.

10. In a gas discharge tube, the factors on which voltage depends are the

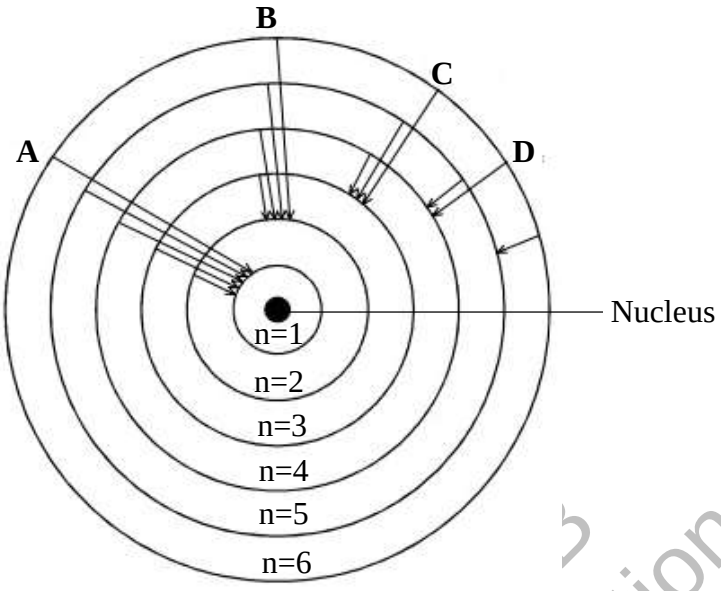
I. length of the tube
II. pressure inside the tube
III. type of gas inside the tube
IV. type of metallic electrodes used

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

Annual Examinations 2024 for Teaching & Learning only
ARD-EB

11. The given image shows the different spectral lines of hydrogen atom.

In the given image, Paschen series is labelled as



12. Which of the following sets of quantum numbers CORRECTLY represents the position of an unpaired electron in 3d orbital?

	n	***l***	**m**	**s**
A	3	1	+2	+½
B	3	1	+1	+½
C	3	2	−1	−½
D	3	2	−3	−½

13. A molecule of water is a polar compound because it

A. can easily dissociate into its ions.
B. can attach to other molecules through hydrogen bonding.
C. has an atom of oxygen that comprises of two lone pair of electrons.
D. has hydrogen and oxygen atoms with difference in electronegativities.

14. The molecular geometry of phosphorus trichloride, PCl_3, is trigonal pyramidal.

According to its shape, how many bond pairs and lone pairs are present in PCl_3?

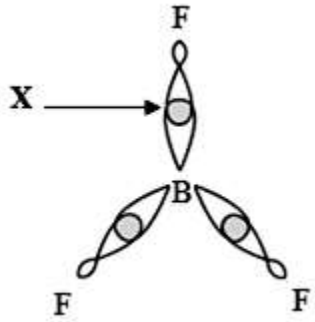
	Number of Bond Pairs	**Number of Lone Pairs**
A	1	3
B	2	1
C	3	0
D	3	1

Annual Examinations 2024 for Teaching & Learning only

15. The type of magnetism depicted by O_2^{2-} is

A. diamagnetic.
B. paramagnetic.
C. ferromagnetic.
D. antiferromagnetic.

16. Consider the given structure of boron trifluoride (BF_3) molecule.



The type of orbital overlap labelled as **X** is

A. sp^3-p
B. sp^2-p
C. sp^3-sp
D. sp^2-sp

17. How many sigma and pi bonds are present in a molecule of C_2H_2?

	Number of Sigma Bonds	**Number of Pi Bonds**
A	2	2
B	2	3
C	3	1
D	3	2

18. Raising the temperature of a gas in a closed system increases the average kinetic energy because the

A. distance between gas molecules decreases.
B. collisions between the molecules increase.
C. intermolecular interactions between molecules increase.
D. collisions between molecules and walls of container decrease.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

19. The maximum density of water occurs at 277 Kelvin and 1 atmospheric pressure.

The approximate temperature in the Fahrenheit scale will be

(**Note**: 0°C = 273.15 K)

A. 4.0°F.
B. 34.2°F.
C. 38.93°F.
D. 550°F.

20. How much pressure in Nm^{-2} is exerted by one mole of a gas enclosed in a vessel of volume 59.4 m^3 at 120°C?

(**Note**: R = 8.314 $JK^{-1}mol^{-1}$)

A. 16.79 Nm^{-2}
B. 21.41 Nm^{-2}
C. 38.21 Nm^{-2}
D. 55.02 Nm^{-2}

21. To avoid a pressure related injury during ascent, a scuba diver must equalise the pressure by

A. breathing normally to add air to their lungs every time they inhale.
B. accumulating air in their mask by breathing out through their nose.
C. releasing extra air from their lungs every time they exhale during breathing.
D. adding air to their ears and sinuses by using one of several ear equalisation techniques.

22. A gas deviates from its ideal behaviour at a greater

A. volume.
B. kinetic energy.
C. compressibility.
D. elastic collision.

23. Surface area is the property of liquids that depends on the

A. viscosity of liquids.
B. capillary action of liquids.
C. cohesive forces between liquid molecules.
D. adhesive forces between molecules and the walls of container.

24. At 4°C, water has a tendency to exhibit maximum

A. weight.
B. density.
C. volume.
D. pressure.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

25. The boiling points of diethyl ether and ethyl alcohol are 34.6ºC and 78.5ºC, respectively.

Comparatively, the low boiling point of diethyl ether indicates that it has

	Volatility	Intermolecular Forces
A	high	weak
B	high	strong
C	low	weak
D	low	strong

26. The amount of energy required to convert one mole of a solid directly into a gas at constant pressure is termed as

A. heat of vaporisation.
B. molar heat of fusion.
C. heat of neutralisation.
D. molar heat of sublimation.

27. Read the given properties.

I. Flows like pure liquids
II. Exhibits symmetry like crystalline solids
III. Shows greater mobility than ordinary liquids
IV. Possesses more organised structure than crystalline solids

The properties that describe a liquid crystal are

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

28. All of the following materials exhibit variations in physical properties along different molecular axes EXCEPT

A. glass.
B. wood.
C. graphite.
D. liquid crystal.

29. In crystallography, if properties of a crystal vary with the change in direction, then this feature is identified as

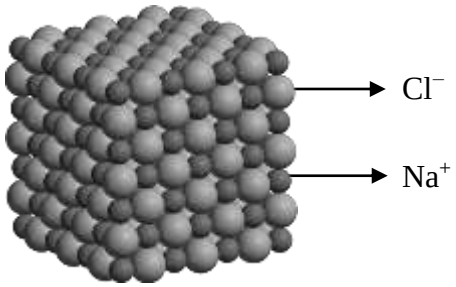
A. allotropy.
B. anisotropy.
C. isomorphism.
D. polymorphism.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

30. All of the following pairs of substances are polymorphs EXCEPT

A. rhombic sulphur and monoclinic sulphur.
B. white phosphorus and red phosphorus.
C. calcium carbonate and sodium nitrate.
D. diamond and graphite.

31. Given is the crystal lattice of sodium chloride.



The MAIN factor(s) that contribute(s) to the lattice energy of this crystal lattice is/ are

I. the size of the ions.
II. the charge on the ions.
III. the size of the crystal.

A. I only.
B. I and II.
C. III only.
D. II and III.

32. The simplest and the least volume consuming repeating structure of any solid is known as a/ an

A. lattice.
B. unit cell.
C. isomorph.
D. polymorph.

33. A TRUE statement about the effect of catalyst is that it

A. provides an alternative route for a reaction.
B. produces high yield of product in an equilibrium.
C. changes the equilibrium constant for the forward reaction.
D. increases the frequency of collisions between reactant particles.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

34. The equilibrium constant K_P for the given reaction can be expressed as

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

$K_P = P_{(CO_2)}$	$K_P = \frac{P_{(CaO)}}{P_{(CaCO_3)}}$
A	B
$K_P = \frac{P_{(CaCO_3)}}{P_{(CaO)} + P_{(CO_2)}}$	$K_P = \frac{P_{(CaO)} + P_{(CO_2)}}{P_{(CaCO_3)}}$
C	D

35. The solubility product (K_{sp}) of $BaSO_4$ is 1.08×10^{-10} at 25°C.

If 10.0 mL solution of 2.0×10^{-3} M Na_2SO_4 is added to 100 mL solution of 3.2×10^{-4} M $BaCl_2$, then $BaSO_4$ will precipitate out because the

A. resultant solution is saturated.
B. resultant solution is unsaturated.
C. value of reaction quotient (Q) is less than K_{sp}.
D. value of reaction quotient (Q) is greater than K_{sp}.

36. Which of the following elements forms an amphoteric as well as normal oxide?

A. Sodium
B. Sulphur
C. Nitrogen
D. Hydrogen

37. If the pH of bleach is 12.6, then the concentration of hydrogen ions in this solution is

A. 2.51×10^{-13} mol/ L
B. 1.4 mol/ L
C. 12.6 mol/ L
D. 3.98×10^{12} mol/ L

38. A basic buffer is prepared by mixing a weak base and a salt of

A. weak acid with a strong acid.
B. weak base with a strong acid.
C. weak acid with a strong base.
D. weak base with a strong base.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

39. Hydrolysis of a salt yields $Ca(OH)_2$ and HCl. The nature of this hydrolysed salt is

A. basic.
B. acidic.
C. neutral.
D. amphoteric.

40. The presence of electric charge on colloidal particles can be observed through

A. ultrafiltration.
B. sedimentation.
C. electrophoresis.
D. Brownian movement.

41. If 2.5 grams of $(NH_2)_2CO$ is dissolved in water to prepare 100 cm^3 solution, the molarity of the solution will be

(**Note**: $^{12}_{6}C$, $^{1}_{1}H$, $^{16}_{8}O$ and $^{14}_{7}N$)

A. 0.025 M.
B. 0.416 M.
C. 4.16 M.
D. 25 M.

42. What happens when a non-volatile solute is added to a pure solvent?

	Escaping Tendency of Solvent Particles	**Vapour Pressure of Solvent**
A	Increases	Decreases
B	Increases	Increases
C	Decreases	Decreases
D	Decreases	Increases

43. Toothpaste is an example of a colloidal solution because its particles

A. can scatter a beam of light.
B. can be seen with the naked eye.
C. can form a homogenous mixture.
D. cannot pass through a filter paper.

44. $H_{2(g)} + Cl_{2(g)} \xrightarrow{hv} 2HCl_{(g)}$

The given reaction is an example of

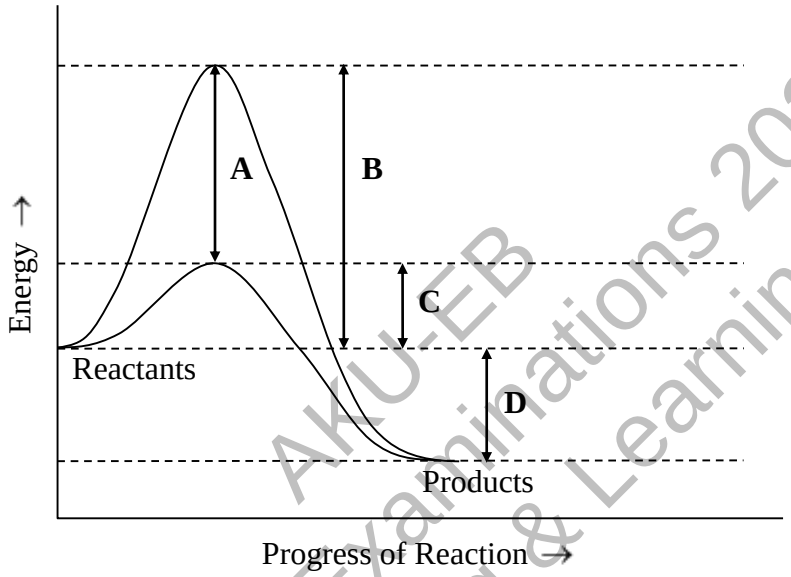
A. zero order reaction.
B. 1st order reaction.
C. 2nd order reaction.
D. 3rd order reaction.

Annual AKU-EB Examinations 2024 for Teaching & Learning only

45. Platinum is used as a catalyst in the manufacturing of sulphuric acid by the contact process. In this process, platinum can become ineffective if it

A. is used in a finely divided form.
B. is employed in a colloidal form.
C. is exposed to high temperatures.
D. reacts with a trace amount of arsenic.

46. The labelled part in the given energy diagram that represents the minimum energy required for the enzyme catalysed reaction is



47. In which of the following reactions does a non-spontaneous reaction occur?

A. Dissolution of ionic salt in a polar solvent
B. Formation of nitric oxide in air by lightning
C. Conversion of liquid water into hydrogen gas
D. Neutralisation of a strong acid by a strong base

48. Mathematically, the first law of thermodynamics is expressed as

A. $\Delta E = \Delta H$
B. $\Delta E = P\Delta V$
C. $\Delta E = q + w$
D. $\Delta E = E_2 - E_1$

49. If the enthalpy of combustion for graphite to form carbon dioxide is –393.7 kJ mol^{-1} and that for carbon monoxide to form carbon dioxide is –283 kJ mol^{-1}, then what will be the enthalpy of formation for carbon monoxide as per the given reaction?

$C_{(s)} + ½\ O_{2(g)} \rightarrow CO_{(g)}$

A. – 676.7 kJ mol^{-1}
B. – 110.7 kJ mol^{-1}
C. + 110.7 kJ mol^{-1}
D. + 676.7 kJ mol^{-1}

AKU-EB Annual Examinations 2024 for Teaching & Learning only

50. An electrical energy of 7.52 J is passed through a calorimeter, increasing its temperature from 0.10 K to 2.5 K.

The heat capacity of the calorimeter will be

A. 2.89 J/K.
B. 3.13 J/K.
C. 18.05 J/K.
D. 19.55 J/K.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only